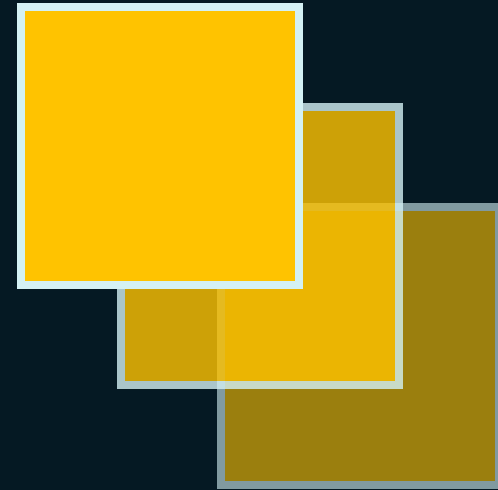




ARRAYS, VECTORS, AND MATRICES IN MACHINE LEARNING

The Foundation of Data
Representation and Processing in AI

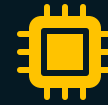


THE BACKBONE OF ML FRAMEWORKS

NumPy, Pandas, TensorFlow, and PyTorch all rely on arrays as their primary data structure for high-performance computing.

Arrays store elements of the same data type in continuous memory blocks, ensuring maximum computational efficiency.

Mastering arrays is the essential first step before advancing to complex vectors and matrices in AI development.



NUMPY



PANDAS



TENSORFLOW



PYTORCH

EFFICIENCY THROUGH BATCH PROCESSING

INDIVIDUAL VARIABLES

```
mark1 = 85  
mark2 = 90  
mark3 = 78  
...  
mark100 = 88
```

Difficult to manage, scale, and process for large datasets.

ARRAY STRUCTURE

```
marks = [  
    85, 90, 78,  
    88, 95  
]
```

85

90

78

88

95

ML BENEFITS

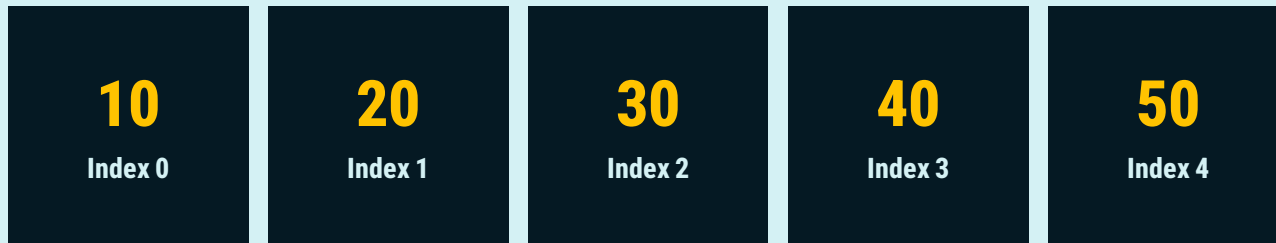
-  Centralized Storage
-  Instant Access
-  Batch Calculations
-  Scalable Processing

1D ARRAYS: SEQUENTIAL DATA STORAGE

A One-Dimensional (1D) array contains a single row of elements. It is the simplest form of structured data used for linear lists.

```
arr = np.array([10, 20, 30, 40, 50])
```

VISUAL REPRESENTATION (MEMORY LAYOUT)



STUDENT MARKS

EMPLOYEE SALARIES

PRODUCT PRICES

TIME SERIES DATA

2D ARRAYS: THE POWER OF GRIDS

2D arrays, or matrices, organize data into a structured format of rows and columns.

They are the standard representation for datasets, Excel sheets, and digital image pixels.

```
arr = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])
```

10	20	30
40	50	60
70	80	90

3X3 MATRIX REPRESENTATION

3D ARRAYS: VOLUMETRIC DATA

A 3D array is conceptually a collection of 2D matrices stacked together to form a data cube.

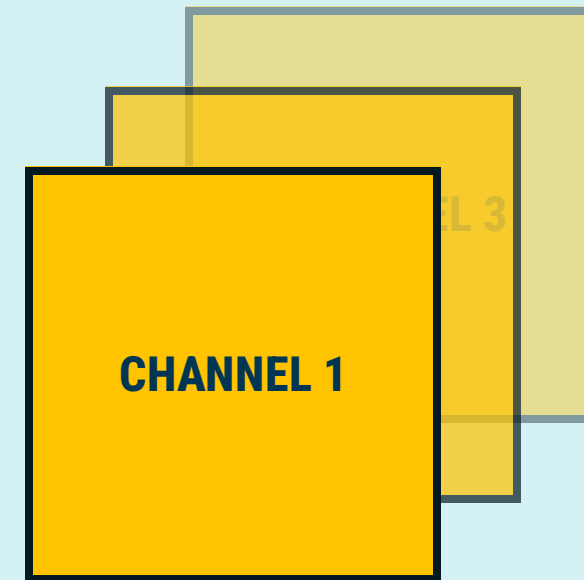
This structure is vital for deep learning tasks involving spatial depth and temporal sequences.

RGB COLOR CHANNELS

VIDEO SEQUENCES

MEDICAL MRI/CT

VOLUMETRIC PHYSICS



DATA CUBE (TENSOR)

VECTORS: INDIVIDUAL OBSERVATIONS

A vector is a 1D array representing one single observation or one set of features.

- ✓ Represents a single data sample
- ✓ Foundation for feature engineering
- ✓ Essential for linear transformations

STUDENT FEATURE VECTOR

STUDY HOURS	6
ATTENDANCE	90
ASSIGNMENTS	8

`np.array([6, 90, 8])`

MATRICES: DATASET CONSOLIDATION

A matrix is a 2D array that stores multiple observations and features in a single computational unit.

In ML, **Rows** represent individual samples (e.g., students), while **Columns** represent specific features (e.g., hours, grades).

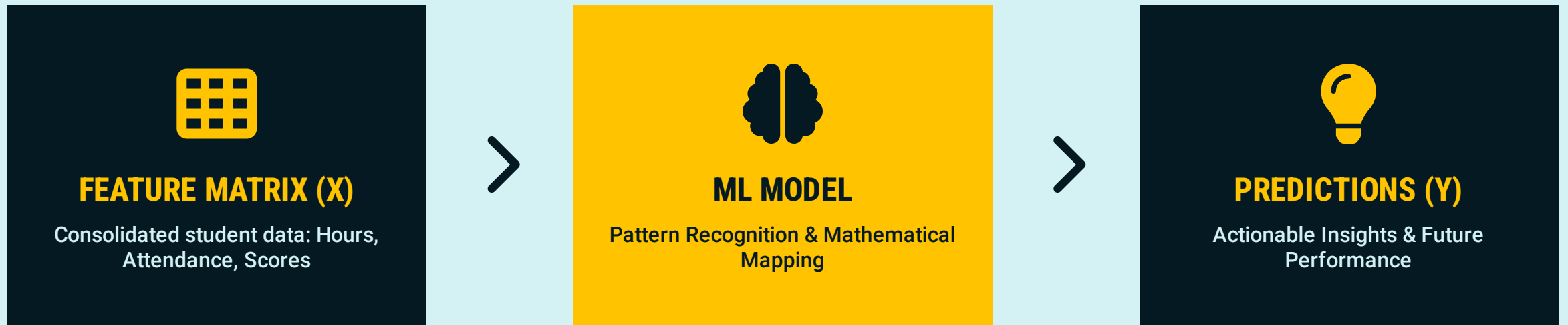
```
students = np.array([
    [6, 90, 8],
    [4, 80, 6],
    [8, 95, 9]
])
```

	Hours	Score	Tasks
	FEATURE MATRIX (X)		
Student A	6	90	8
Student B	4	80	6
Student C	8	95	9

COMPARATIVE ANALYSIS

FEATURE	ARRAY	VECTOR	MATRIX
DEFINITION	General collection of elements	1D specialized array	2D specialized array
DIMENSIONS	N-Dimensional (1D, 2D, 3D...)	Strictly 1D	Strictly 2D
EXAMPLE	[10, 20, 30]	[6, 90, 8]	[[6, 90, 8], [4, 80, 6]]
ML USE	Generic data storage	Represents one single sample	Represents a complete dataset

TRANSFORMING DATA INTO INSIGHTS



Machine Learning models identify hidden patterns within feature matrices to learn relationships and make accurate predictions on new, unseen data.

THE PATH TO AI MASTERY

ARRAYS

The Foundation (Generic Storage)

VECTORS

Individual Records (1D)

MATRICES

Complete Datasets (2D)

TENSORS

Multi-dimensional AI Data

Mastering these structures is the first step toward building efficient, scalable, and powerful Machine Learning models.

FINISH